MODULE - II

Series Solution & Special Functions

Series Solution of Second Order Differential Equations

The solution of ordinary linear differential equation of second order with variable coefficients in the form of an infinite convergent series is called Series solution of second order differential equation.

For Example :-

Let the second order differential equation be,

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = 0$$

$$\Rightarrow (D^2 - 2D + 1)\, y = 0$$

$$\Rightarrow (D-1)^2 = 0$$

$$\Rightarrow D = 1, 1$$

$$y = (C_1 + xC_2)e^x$$

The expansion is,

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \ldots$$

$$xe^x = x + \frac{x^2}{1!} + \frac{x^3}{2!} + \ldots$$

$$\therefore y = C_1\left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \ldots\right) + C_2\left(x + \frac{x^2}{1!} + \ldots\right)$$

Ordinary Point

A point $x = a$ is called an ordinary point of the differential equation of second order,

$$\frac{d^2y}{dx^2} + P(x)\frac{dy}{dx} + Q(x)y = 0$$

where, $P(x)$ & $Q(x)$ are polynomial, if both $P(x)$ & $Q(x) \not\to \infty$ at $x = a$.

Question: Show that $x = 0$ is an ordinary point for the differential equation,

$$(1+x^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx} - y = 0$$

Sol^n: The differential eq^n. is,

$$(1+x^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx} - y = 0$$

$$\Rightarrow \frac{d^2y}{dx^2} + \frac{x}{(1+x^2)} \cdot \frac{dy}{dx} - \frac{1}{(1+x^2)} y = 0$$

Comparing above eqn. with standard form, we get,

$$P(x) = \frac{x}{1+x^2} \quad \& \quad Q(x) = \frac{-1}{1+x^2}$$

At $x=0$, $P(x)=0$ & $Q(x)=-1$

Since, both $P(x)$ & $Q(x) \not\to \infty$ at $x=0$.

$\therefore$ $x=0$ is an ordinary point for the given second order differential equation.

Singular Point

A point $x=a$ is said to be singular point of the second order differential equation,

$$\frac{d^2y}{dx^2} + P(x)\frac{dy}{dx} + Q(x)y = 0,$$

if at least one of $P(x)$ & $Q(x) \to \infty$ at $x=a$.

There are two types of singular point:

1. Regular Singular Point:
A singular point $x=a$ is said to be regular singular point of second order differential equation,

$$\frac{d^2y}{dx^2} + P(x)\frac{dy}{dx} + Q(x)y = 0$$

if both $(x-a)P(x)$ & $(x-a)^2Q(x) \not\to \infty$ at $x=a$.

2. Irregular Singular Point:
A singular point which is not regular is known as irregular singular point.

Question: Show that $x=0$ is a regular singular point of $2x^2y'' + 3xy' + (x^2-4)y = 0$.

Sol^n:

The given differential eq^n. is,

$$2x^2y'' + 3xy' + (x^2-4)y = 0$$

$$\Rightarrow y'' + \frac{3}{2x}y' + \frac{x^2-4}{2x^2}\cdot y = 0$$

Comparing with standard form, we get,

$$P(x) = \frac{3}{2x} \quad \& \quad Q(x) = \frac{x^2-4}{2x^2}$$

Now,

$$(x-a)P(x) = xP(x) = \frac{3}{2}$$

$$(x-a)^2 Q(x) = x^2 Q(x) = -2$$

Since, both does not tends to infinity.

$\therefore$ $x=0$ is a regular singular point of given equation.